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Batch-4

LSTM Practice Questions — Answers

Question 1: Numerical — Single-Cell Forward Pass

Given:

Input = 0.5, Previous hidden state = 0.2, Previous cell state = 0.4.

1. Forget Gate

ft = sigmoid(0.7 × 0.5 + 0.3 × 0.2 + 0.1)

ft = sigmoid(0.35 + 0.06 + 0.1)

ft = sigmoid(0.51)

ft = 1 / (1 + e^(-0.51))

ft = 0.6248

2. Input Gate

it = sigmoid(0.6 × 0.5 + 0.4 × 0.2 + 0)

it = sigmoid(0.30 + 0.08)

it = sigmoid(0.38)

it = 0.5939

3. Candidate Cell State

$\tilde{C}t$ = tanh(0.9 × 0.5 + 0.2 × 0.2)

$\tilde{C}t$ = tanh(0.45 + 0.04)

$\tilde{C}t$ = tanh(0.49)

$\tilde{C}t$ = 0.4542

4. New Cell State

Ct = (0.6248 × 0.4) + (0.5939 × 0.4542)

Ct = 0.2499 + 0.2698

Ct = 0.5197

5. Output Gate

ot = sigmoid(0.5 × 0.5 + 0.5 × 0.2 + 0.1)

ot = sigmoid(0.25 + 0.10 + 0.10)

ot = sigmoid(0.45)

ot = 0.6106

6. New Hidden State

ht = 0.6106 × tanh(0.5197)

ht = 0.6106 × 0.4775

ht = 0.2915

Final answers:

Forget gate = 0.6248

Input gate = 0.5939

Candidate cell state = 0.4542

New cell state = 0.5197

Output gate = 0.6106

New hidden state = 0.2915

Question 2: Numerical — LSTM Parameter Count

Given:

Input dimension = 10

Hidden units = 32

Number of gates = 4

(a) Standard formulation with one bias vector per gate

Input-to-hidden parameters = 32 × 10 = 320

Hidden-to-hidden parameters = 32 × 32 = 1024

Bias parameters = 4 × 32 = 128

Total parameters = 320 + 1024 + 128 + 320 + 1024 + 128?

Total parameters = 4 × (320 + 1024 + 32)

Total parameters = 4 × 1376

Total trainable parameters = 5504

(b) PyTorch formulation with two bias vectors per gate

Input-to-hidden parameters = 4 × 32 × 10 = 1280

Hidden-to-hidden parameters = 4 × 32 × 32 = 4096

Bias parameters = 4 × 2 × 32 = 256

Total parameters = 1280 + 4096 + 256

Total trainable parameters = 5632

Question 3: Numerical — Vanishing Gradient Through the Cell State

(a) Scaling factor for forget gate = 0.9

After 10 steps:

0.9 × 0.9 × 0.9 × 0.9 × 0.9 × 0.9 × 0.9 × 0.9 × 0.9 × 0.9

Scaling factor = 0.3487

After 50 steps:

0.9 raised to the power of 50

Scaling factor = 0.0052

(b) Scaling factors after 50 steps

For forget gate = 0.5:

0.5 raised to the power of 50

Scaling factor = 8.8818×10^{-16}

For forget gate = 0.99:

0.99 raised to the power of 50

Scaling factor = 0.6050

(c) Explanation

These results show that the gradient becomes very small when information passes through many time steps, especially when the forget-gate value is low. With a forget-gate value of 0.99, much more of the gradient is retained after 50 steps. LSTMs reduce

the vanishing-gradient problem through their cell state and gating mechanisms, allowing them to learn long-term dependencies more effectively than vanilla RNNs.

Question 4: Real-World Scenario — Electricity Load Forecasting

Given:

Duration = 3 years

Days per year = 365

Hours per day = 24

Lookback window = 168 hours

Prediction horizon = 24 hours

Features per hour = 3

(a) Total number of hourly records

Total records = 3 × 365 × 24

Total records = 3 × 8760

Total records = 26280

(b) Number of input/output windows

Total windows = 26280 − 168 − 24 + 1

Total windows = 26280 − 192 + 1

Total windows = 26089

(c) Tensor shapes

Batch size = 64

Lookback window = 168

Number of features = 3

Input shape = (64, 168, 3)

The model predicts 24 hourly load values for each window.

Output shape = (64, 24)

If the output includes a separate dimension for the single predicted feature, its shape is (64, 24, 1).

(d) Training windows and mini-batches

Training records = 26280 × 0.70

Training records = 18396

Training windows = 18396 − 168 − 24 + 1

Training windows = 18396 − 192 + 1

Training windows = 18205

Number of mini-batches = 18205 ÷ 64

Number of mini-batches = 284.453125

Rounding up to include the final partial batch:

Mini-batches per epoch = 285

The final partial batch contains 29 windows.

(e) Trainable parameters for hidden size 64

Input dimension = 3

Hidden size = 64

Number of gates = 4

Input-to-hidden parameters = 4 × 64 × 3 = 768

Hidden-to-hidden parameters = 4 × 64 × 64 = 16384

Bias parameters = 4 × 64 = 256

Total parameters = 768 + 16384 + 256

Total trainable parameters = 17408

Question 5: Real-World Scenario — Predictive Maintenance

(a) Min-max normalisation

Minimum reading = 20 mm/s

Maximum reading = 120 mm/s

Actual reading = 95 mm/s

Normalised value = (95 − 20) ÷ (120 − 20)

Normalised value = 75 ÷ 100

Normalised value = 0.75

(b) Failure prediction

Predicted failure probability = 0.83

Decision threshold = 0.5

Since 0.83 is greater than 0.5, the model predicts a failure.

Prediction = Failure

(c) Confusion matrix and performance metrics

Total operating cycles = 1000

Actual failures = 40

Predicted failures = 45

Correctly predicted failures = 30

True Positives (TP) = 30

False Positives (FP) = 45 – 30 = 15

False Negatives (FN) = 40 – 30 = 10

True Negatives (TN) = 1000 – 30 – 15 – 10 = 945

Confusion matrix:

	Actual Failure	**Actual No Failure**
Predicted Failure	30	15
Predicted No Failure	10	945

Precision

Precision = 30 ÷ (30 + 15)

Precision = 30 ÷ 45

Precision = 66.67%

Recall

Recall = 30 ÷ (30 + 10)

Recall = 30 ÷ 40

Recall = 75%

F1-score

F1-score = 2 × (0.6667 × 0.75) ÷ (0.6667 + 0.75)

F1-score = 1.00005 ÷ 1.4167

F1-score ≈ 70.59%

Accuracy

Accuracy = (30 + 945) ÷ 1000

Accuracy = 975 ÷ 1000

Accuracy = 97.5%

Final results:

True Positives = 30

False Positives = 15

False Negatives = 10

True Negatives = 945

Precision = 66.67%

Recall = 75%

F1-score = 70.59%

Accuracy = 97.5%

(d) Which metric should the maintenance team prioritise?

I would prioritise recall because missing an actual pump failure can cause unexpected downtime, equipment damage, and expensive repairs. Higher recall means that the model identifies more of the pumps that are actually going to fail. Accuracy can be misleading because only 40 out of 1000 cycles result in actual failures. A model can achieve high accuracy by predicting that most cycles will not fail while still missing important failures. Precision should also be monitored to avoid unnecessary maintenance caused by false alarms.

LSTM cell has a memory of 4 from the past
Previous Hidden State 0.5 at the current time step
it processes new input token:
Forget Gate Activation = 0.7
Input Gate Activation = 0.5
New Candidate Memory = 2.4
Output Gate = 0.6
Updated Cell State?
&
Final Hidden State?
Hint: tanh (4.0) ≈ 1

$C_{t-1} = 4$
$f_t = 0.7$
$i_t = 0.5$
$\tilde{C}_t = 2.4$
$o_t = 0.6$

$C_t = f_t C_{t-1} + i_t \tilde{C}_t$

$C_t = 0.7(4) + 0.5(2.4)$
$= 2.8 + 1.2 = 4$

$h_t = o_t \tanh(C_t)$
$= 0.6 \tanh(4.0)$
$= 0.6 (1) = 0.6$

Updated Cell State = 4
Final Hidden State = 0.6